

Does Unification Help? The Intermittency Parameter and Joint SPX/VIX Calibration

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Abstract

Pure rough Heston/Bergomi models are known to struggle with joint calibration to SPX and VIX option smiles simultaneously. We ask whether the log S-fBM model’s extra intermittency parameter, λ^2 , offers a structural fix. Working directly from the model’s closed-form covariance, we prove that λ^2 cancels out of the ratio governing short-dated skew versus VIX-level vol-of-vol at fixed H , and is therefore calibration-inert for this purpose; the correlation-timescale parameter T is the actual lever. We verify this cancellation numerically to machine precision. We then show, via a cross-paper derivation, that the same aggregation mechanism explaining index-versus-single-name roughness differences likely undermines rather than reinforces this T -lever at the index level – a genuine tension between two of this project’s most promising theoretical threads, not a synergy.

1 Background and motivation

[Gatheral et al. \[2020\]](#) precisely characterize the joint SPX/VIX calibration failure of pure rough volatility models: matching the steep, near-divergent short-maturity SPX skew requires a level of vol-of-vol inconsistent with observed VIX-implied-volatility levels. This sits within the broader “Guyon conjecture” territory [[Guyon, 2022](#)] that continuous-price-path models may structurally struggle to fit both surfaces at once. Known fixes – quadratic rough Heston [[Gatheral et al., 2020](#)], path-dependent volatility [[Guyon and Lekeufack, 2023](#), [Gazzani and Guyon, 2025](#)], Gaussian polynomial Volterra models [[Abi Jaber and Li, 2025](#)] – all add structure beyond plain fBM-driven log-volatility.

Finding 1.1. No paper tests log S-fBM, or any structurally similar unified model, against the joint SPX/VIX calibration problem. We confirmed this independently via two separate literature searches. The source paper itself [[Wu et al., 2022](#)] is purely a probability/statistics paper with no options-pricing content; the authors explicitly flag “a faithful model for asset and option prices” as future work.

[Abi Jaber et al. \[2025\]](#) provide a relevant cautionary precedent: a conventional one-factor Markovian model *outperforms* rough and path-dependent alternatives with the same parameter count on this exact benchmark. Extra structural sophistication does not automatically buy better joint fit.

2 A derived negative result: λ^2 cancels out

Recall the log S-fBM covariance (Wu et al., 2022, verified directly from the source):

$$C(\tau; H, T, \nu^2) = \frac{\nu^2}{2} [T^{2H} - \tau^{2H}], \quad \nu^2 = \frac{\lambda^2}{H(1-2H)}. \quad (1)$$

Theorem 2.1. *For fixed H and T , the ratio $R(\tau_1, \tau_2) \equiv C(\tau_1; H, T, \nu^2)/C(\tau_2; H, T, \nu^2)$ is independent of λ^2 for all $\tau_1, \tau_2 < T$.*

Proof. ν^2 enters (1) as a common multiplicative prefactor: $C(\tau; H, T, \nu^2) = \nu^2 \cdot g(\tau; H, T)$ where $g(\tau; H, T) = \frac{1}{2}[T^{2H} - \tau^{2H}]$ does not depend on ν^2 (equivalently, on λ^2). Hence $R(\tau_1, \tau_2) = g(\tau_1; H, T)/g(\tau_2; H, T)$, in which ν^2 – and therefore λ^2 – does not appear. $\square$

Taking $\tau_1 = \tau_{\text{short}}$ (a short SPX maturity) and $\tau_2 = \tau_{\text{VIX}}$ ($\approx$ one month), Theorem 1 says the ratio between short-dated and VIX-horizon vol-of-vol is entirely controlled by H and T , and *not at all* by λ^2 – for any fixed H , no choice of intermittency can differentially target short-dated skew steepness relative to the VIX level.

Corollary 2.2. *T is the parameter that governs $R(\tau_{\text{short}}, \tau_{\text{VIX}})$: as $T \rightarrow \infty$ (the pure-fBM limit), $R \rightarrow 1$ and the lever vanishes; as $T \rightarrow \tau_{\text{VIX}}^+$, $R \rightarrow \infty$, its maximal differentiating power.*

This makes T – not λ^2 – structurally analogous to the second, slower timescale already used in existing multi-factor fixes for the joint calibration problem [Gazzani and Guyon, 2025].

3 Numerical verification

We evaluated (1) directly at $H = 0.10$, $T = 0.5$ years, $\tau_{\text{short}} = 1/252$, $\tau_{\text{VIX}} = 21/252$.

λ^2	0.01	0.05	0.10	0.50	2.00
$R(\tau_{\text{short}}, \tau_{\text{VIX}})$	2.058205	2.058205	2.058205	2.058205	2.058205

Table 1: The ratio is invariant to λ^2 to six decimal places across a $200\times$ range, confirming Theorem 1 exactly.

T	0.10	0.25	0.50	1.00	2.00	5.00	20.0	10^5
R	13.28	2.86	2.06	1.71	1.51	1.36	1.23	1.03

Table 2: $R(\tau_{\text{short}}, \tau_{\text{VIX}})$ is strictly monotone decreasing in T , confirming Corollary 1.

Finding 3.1. Theorem 1 and Corollary 1 are now verified facts, not merely derived claims: the cancellation is exact to numerical precision, and the T -dependence is exactly as predicted in both limiting regimes.

4 A cross-paper tension: does the T -lever survive aggregation?

The companion paper on aggregation (this series, Paper 2) shows that cross-sectional averaging over heterogeneous single-name log-vol processes pulls the index’s Hurst exponent toward that of a smoother common factor once $\beta^4 N_s \gg 10$. We ask whether the same mechanism affects the index’s effective T .

Proposition 4.1. *Under the additive factor model $\omega_i(t) = \beta_i F(t) + X_i(t)$ with independent components, the index covariance is*

$$\text{Cov}(\Omega) = \left(\sum_i w_i \beta_i \right)^2 C_F(\tau) + \left(\sum_i w_i^2 \right) C_X(\tau).$$

As N_s grows, $\sum_i w_i^2 \rightarrow 0$ (equal-weighted case, at rate N_s^{-1} or faster under the diversification scaling of Paper 2), so $\text{Cov}(\Omega) \rightarrow (\sum_i w_i \beta_i)^2 C_F(\tau)$ pointwise in τ – the entire covariance profile, truncation range included, converges to the common factor’s shape, not merely its short-lag behavior.

If $T_F > T_X$ – plausible if market-wide volatility regimes persist longer than firm-specific idiosyncratic shocks, though this is an economic premise we do not establish – then the index’s effective T is pulled toward the common factor’s (longer) value under the same mechanism that raises its effective H .

Finding 4.2. Combining this with Corollary 1: $R(\tau_{\text{short}}, \tau_{\text{VIX}})$ is strictly decreasing in T , so a longer effective index-level T weakens the very lever needed to decouple SPX skew from VIX level. Since SPX is itself an aggregate over roughly 500 names, the diversification mechanism that would make it a well-behaved index (Paper 2) simultaneously erodes the mechanism proposed here to fix its joint calibration problem. This is a genuine, non-forced tension between two of this project’s most promising theoretical threads – not a synergy – and it generates a new falsifiable prediction: T_{index} should open up specifically once $\beta^4 N_s \gtrsim 10$, the same threshold governing H_{index} .

5 Findings summary

- **Established.** The joint SPX/VIX calibration problem is real and precisely characterized [Gatheral et al., 2020].
- **Negative result.** λ^2 cannot be the fix (Theorem 1), verified numerically to machine precision.
- **Plausible but unverified.** T is a structurally sensible lever (Corollary 1), analogous to existing multi-factor fixes, but untested against real option data.
- **Plausible but unverified.** The aggregation-driven erosion of the T -lever at index level (Finding 4.2), conditional on the unestablished premise $T_F > T_X$.
- **Speculative/gap.** Whether unification helps joint calibration at all remains untested; no pricing engine for log S-fBM currently exists.

6 Future directions

1. Build a minimal log S-fBM option-pricing/calibration engine (likely simulation-based, since the model is not obviously affine or Markovian) and test the T -lever hypothesis directly against public SPX/VIX option-chain data.

2. Include an explicit parameter-matched overfitting control, following the precedent of [Abi Jaber et al. \[2025\]](#), so that any apparent improvement cannot be attributed merely to an extra free parameter.
3. Test the new prediction from Finding 4.2 directly: estimate T separately for single names and the index across sub-portfolios of varying N_s and mean β , checking whether T_{index} opens up at the same threshold that governs H_{index} .
4. If the premise $T_F > T_X$ fails empirically, reassess whether a genuinely index-level (not inherited via aggregation) fast timescale could be sourced separately – e.g. from market-wide liquidity or leverage-driven vol-of-vol – to preserve the T -lever’s usefulness despite the aggregation tension.

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